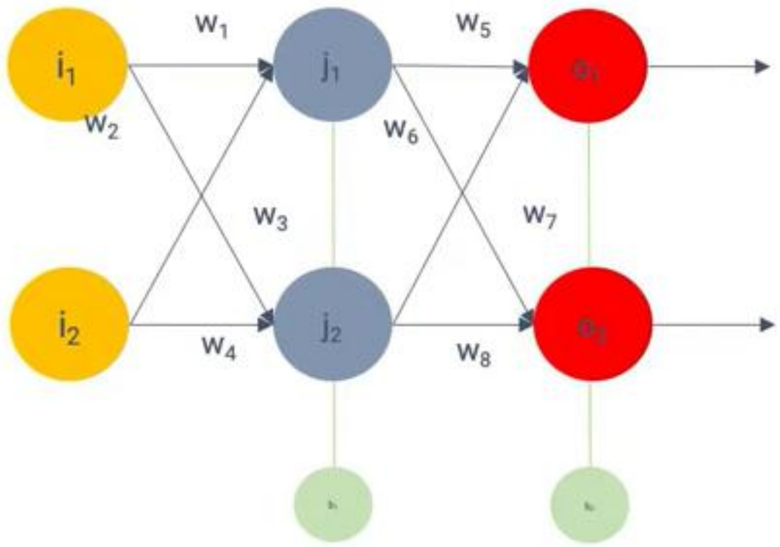
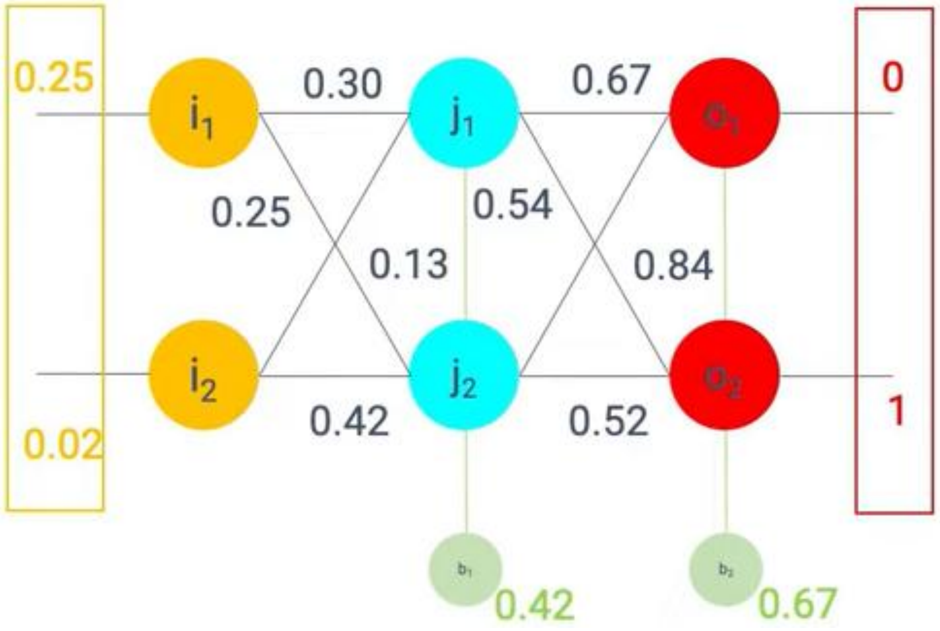
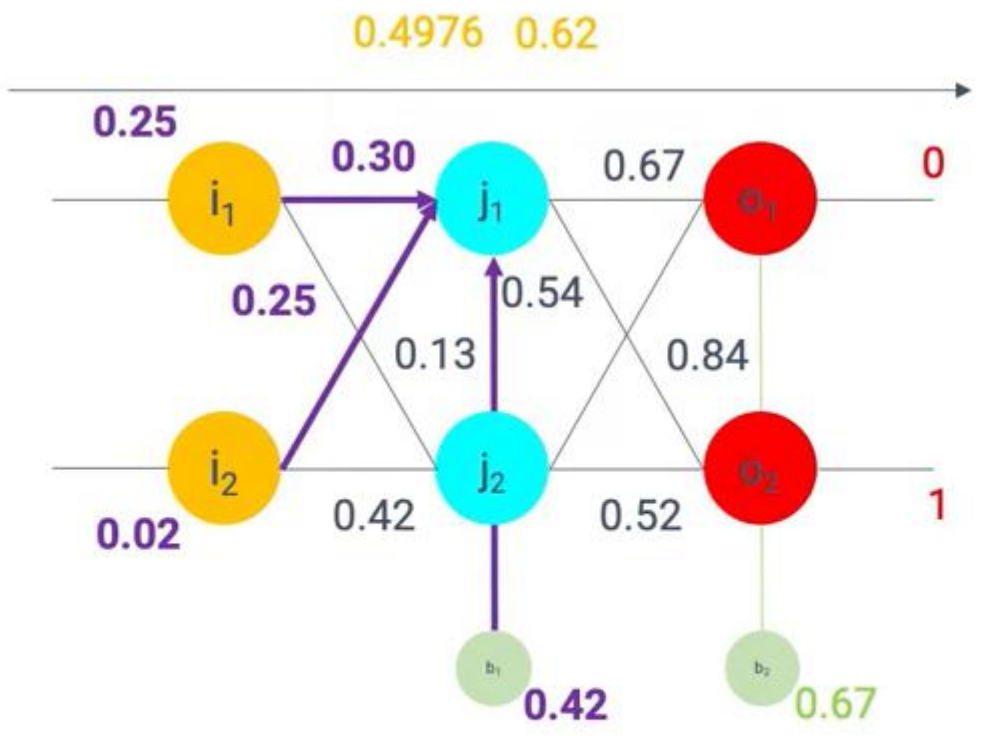
Backpropagation



Initialization



Forward propagation



Compute of j_1

$$j_1 = i_1 \cdot w_1 + i_2 \cdot w_2 + b_1$$

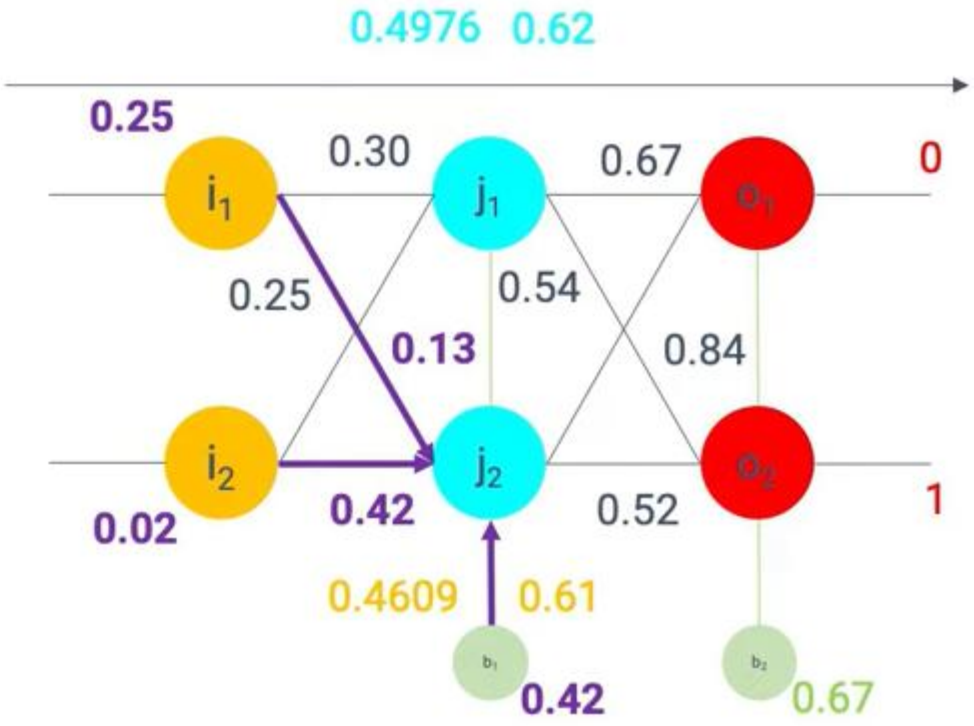
$$j_1 = 0.25 \cdot 0.30 + 0.02 \cdot 0.25 + 0.42$$

$$j_1 = 0.075 + 0.005 + 0.42$$

$$j_1 = 0.5$$

$$f(j_1) = \frac{1}{1+e^{(-0.5)}} = 0.62$$

Forward propagation



Compute of j_2

$$j_2 = i_1 \cdot w_3 + i_2 \cdot w_4 + b_1$$

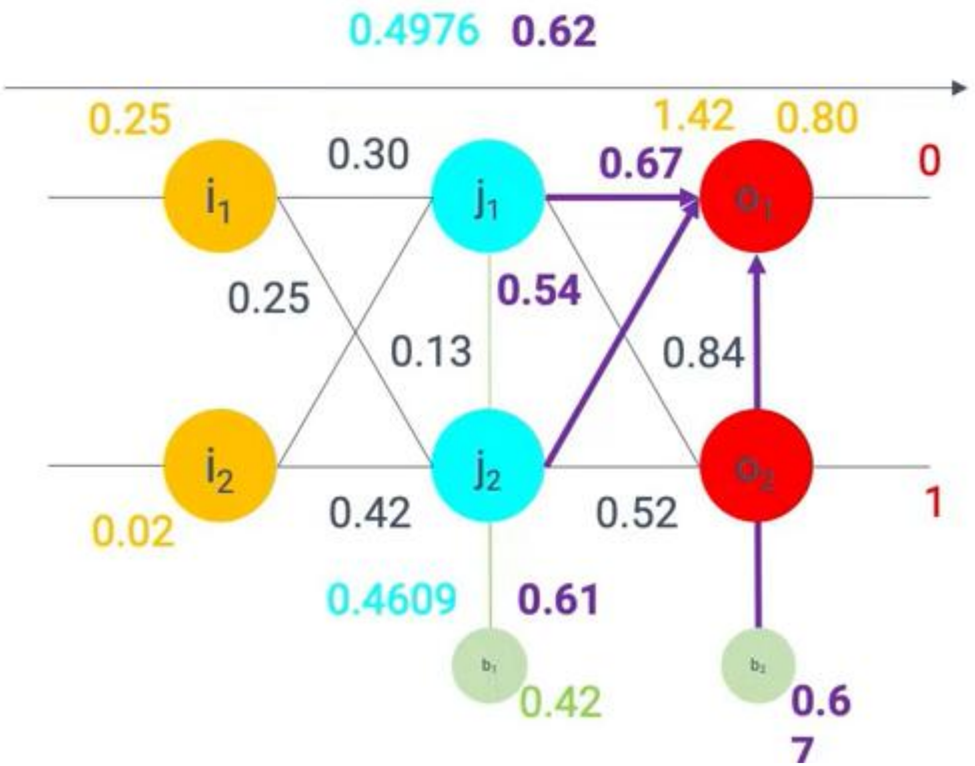
$$j_2 = 0.25 \cdot 0.13 + 0.02 \cdot 0.42 + 0.42$$

$$j_2 = 0.0325 + 0.0086 + 0.42$$

$$j_2 = 0.4609$$

$$f(j_2) = \frac{1}{1+e^{(-0.4609)}} = 0.61$$

Forward propagation



Compute of o_1

$$o_1 = f(j_1) \cdot w_5 + f(j_2) \cdot w_6 + b_2$$

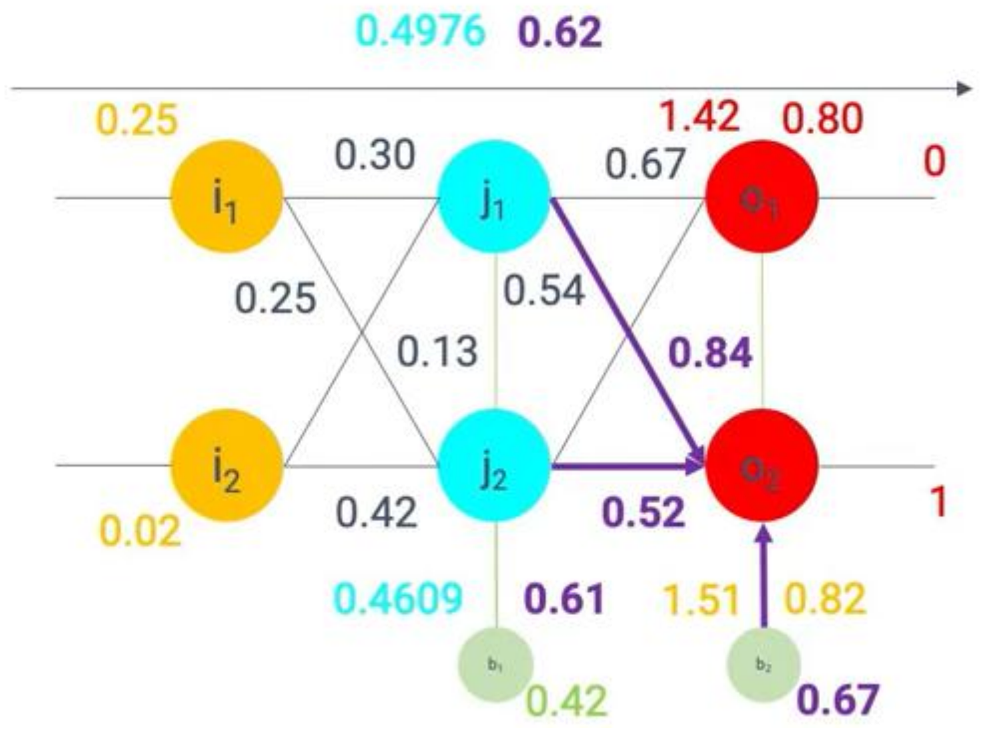
$$o_1 = 0.62 \cdot 0.67 + 0.61 \cdot 0.54 + 0.67$$

$$o_1 = 0.42 + 0.33 + 0.67$$

$$o_1 = 1.42$$

$$f(o_1) = \frac{1}{1+e^{(-1.42)}} = 0.80$$

Forward propagation



Compute of o_2

$$o_2 = f(j_1) \cdot w_5 + f(j_2) \cdot w_6 + b_2$$

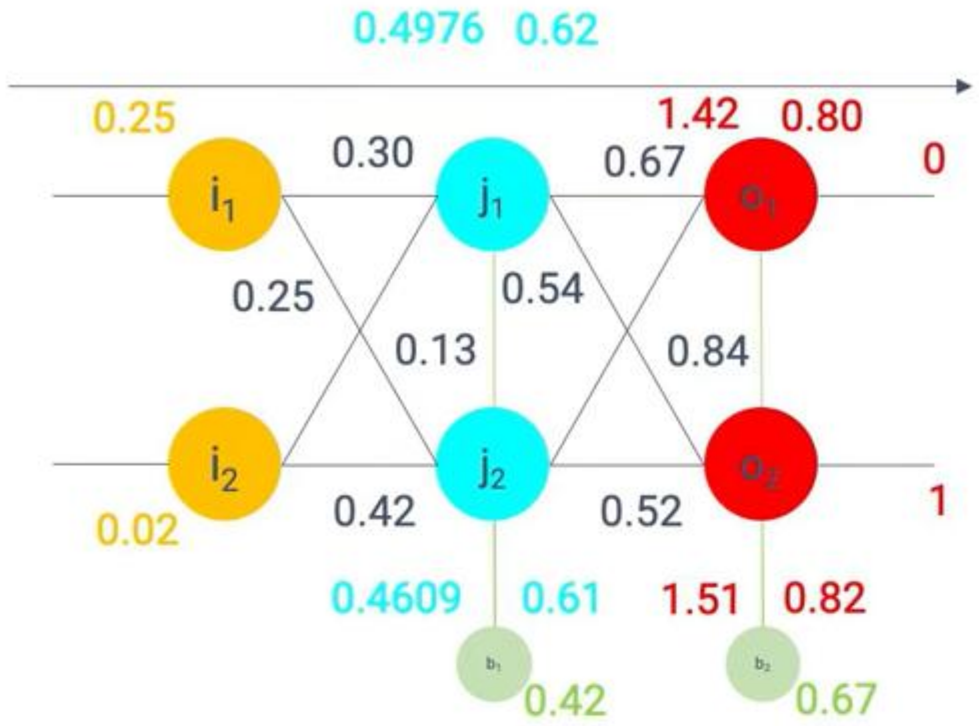
$$o_2 = 0.62 \cdot 0.84 + 0.61 \cdot 0.52 + 0.67$$

$$o_2 = 0.52 + 0.32 + 0.67$$

$$o_2 = 1.51$$

$$f(o_2) = \frac{1}{1+e^{(-1.51)}} = 0.82$$

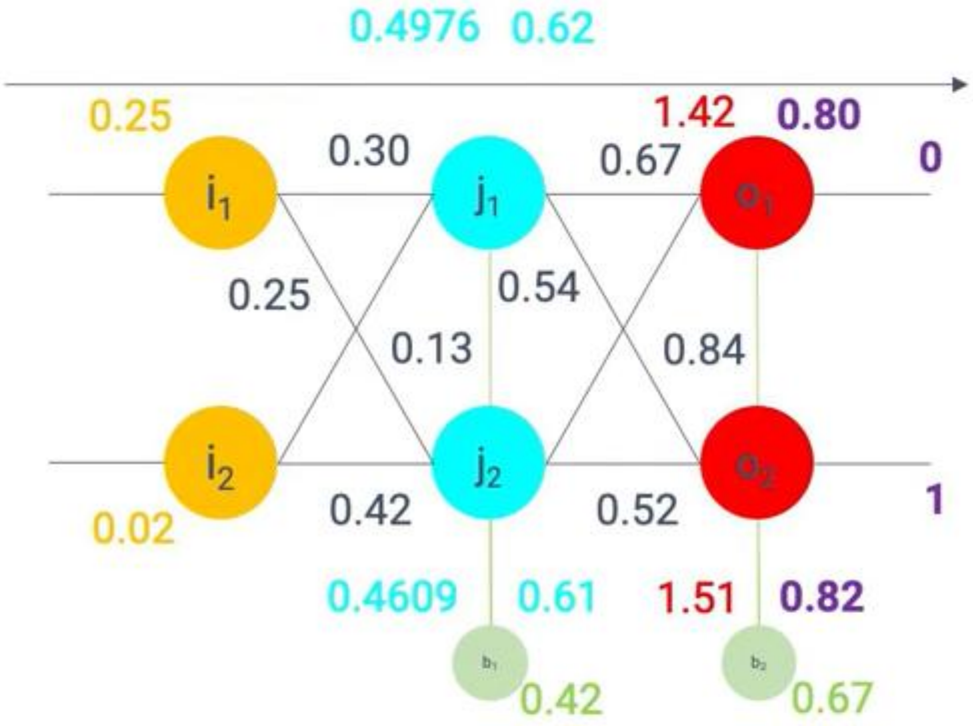
Calculate the error



For each neuron:

$$E_{total} = \sum \frac{1}{2}(target - actual)^2$$

Calculate the error



For each neurone

$$E_{neurone} = \sum \frac{1}{2}(target - actual)^2$$

$$E_{o1} = \frac{1}{2}(target_{o1} - actual_{o1})^2$$

$$E_{o1} = \frac{1}{2}(0 - 0.80)^2 = 0.32$$

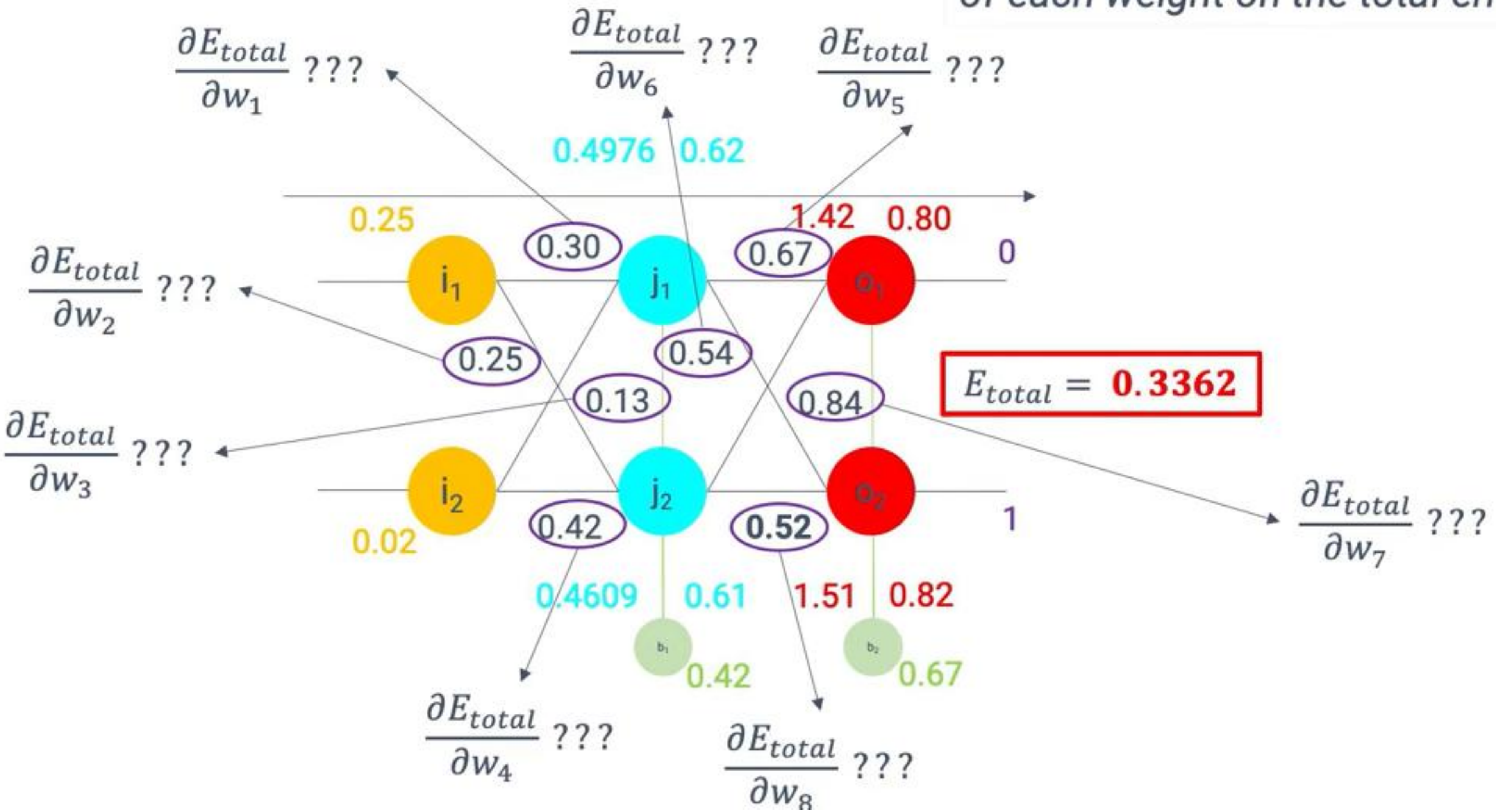
$$E_{o2} = \frac{1}{2}(target_{o2} - actual_{o2})^2$$

$$E_{o2} = \frac{1}{2}(1 - 0.82)^2 = 0.0162$$

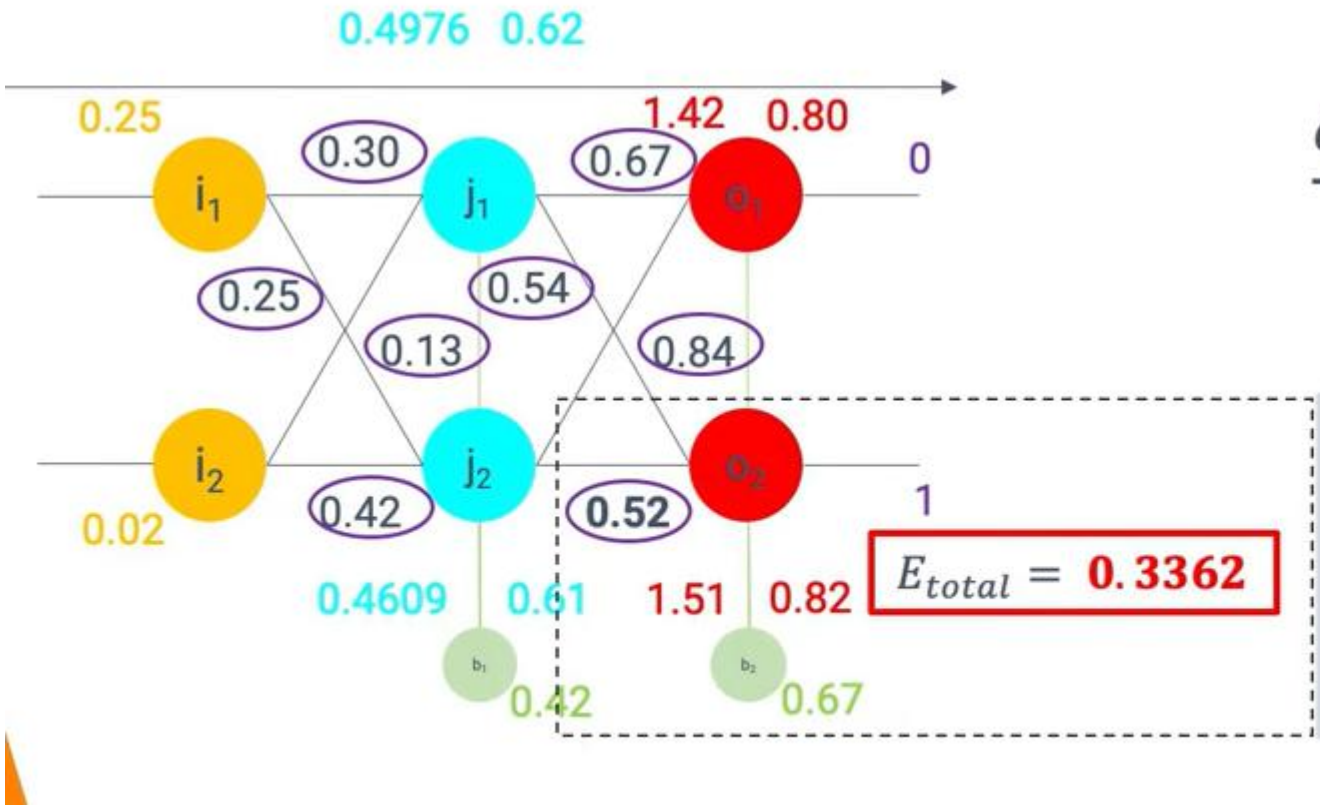
$$E_{total} = E_{o1} + E_{o2} = 0.32 + 0.0162 = \mathbf{0.3362}$$

Backward propagation

We are looking to measure the impact of each weight on the total error.

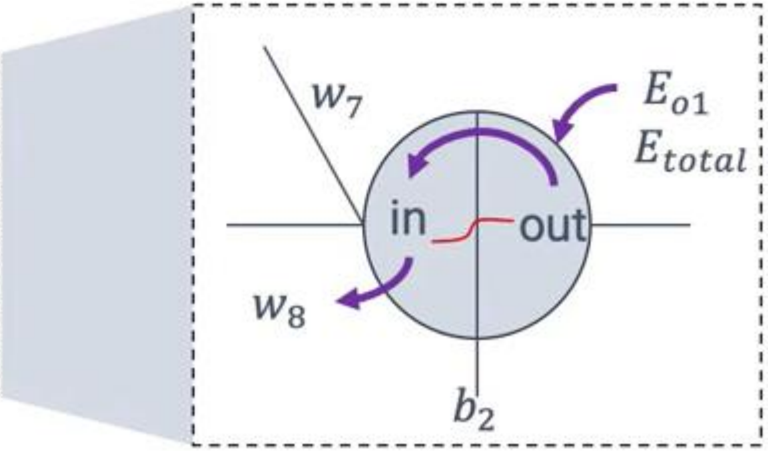


Backward propagation

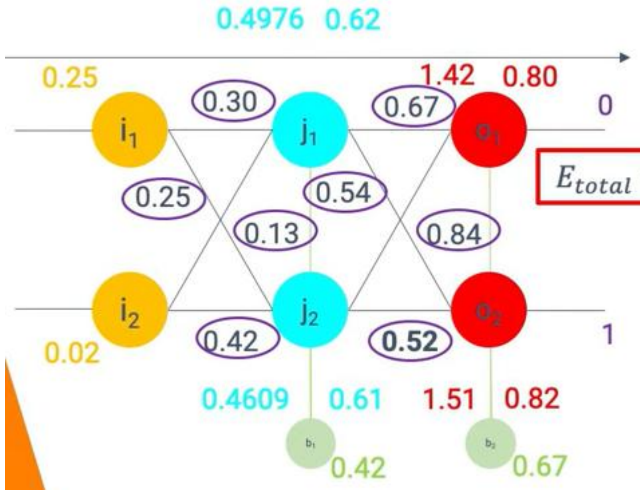


Adjust weights for output layer

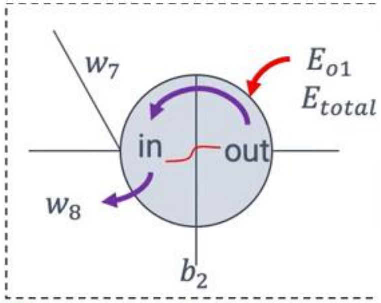
$$\frac{\partial E_{total}}{\partial w_8} = \frac{\partial E_{total}}{\partial out_{o1}} \cdot \frac{\partial out_{o1}}{\partial in_{o1}} \cdot \frac{\partial in_{o1}}{\partial w_8}$$



Backward propagation



$\boxed{E_{total} = \mathbf{0.3362}}$



$$\frac{\partial E_{total}}{\partial w_8} = \boxed{\frac{\partial E_{total}}{\partial out_{o2}}} \cdot \frac{\partial out_{o2}}{\partial in_{o2}} \cdot \frac{\partial in_{o2}}{\partial w_8}$$

$$E_{total} = E_{o1} + E_{o2}$$

$$E_{o1} = \frac{1}{2}(target_{o1} - out_{o1})^2$$

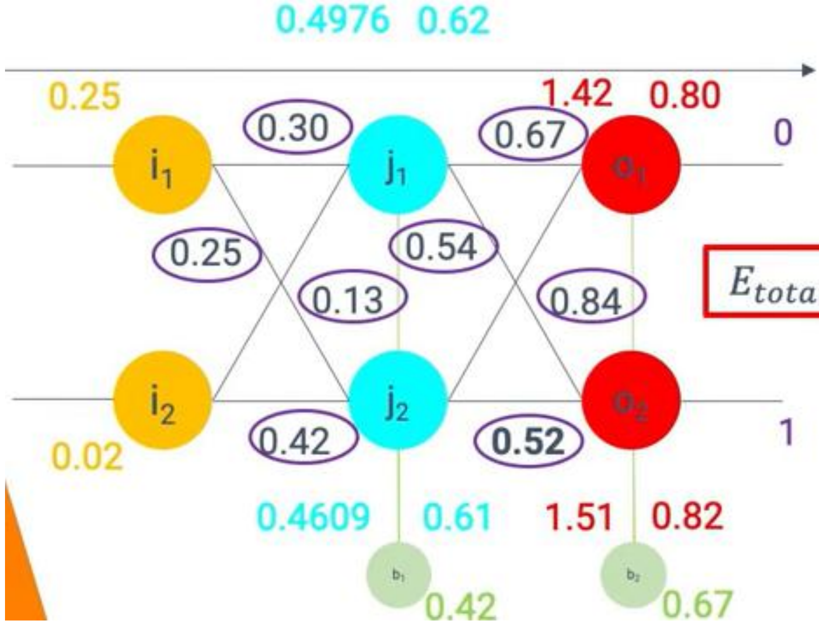
$$E_{o2} = \frac{1}{2}(target_{o2} - out_{o2})^2$$

$$E_{total} = \frac{1}{2}(target_{o1} - out_{o1})^2 + \frac{1}{2}(target_{o2} - out_{o2})^2$$

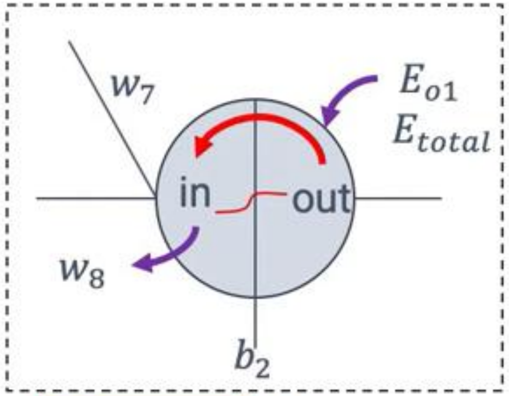
$$\frac{\partial E_{total}}{\partial out_{o2}} = 0 + 2 \cdot \frac{1}{2}(target_{o2} - out_{o2})^{2-1} \cdot -1$$

$$\frac{\partial E_{total}}{\partial out_{o2}} = out_{o2} - target_{o2} = 0.82 - 1 = -0.18$$

Backward propagation



$E_{total} = \mathbf{0.3362}$

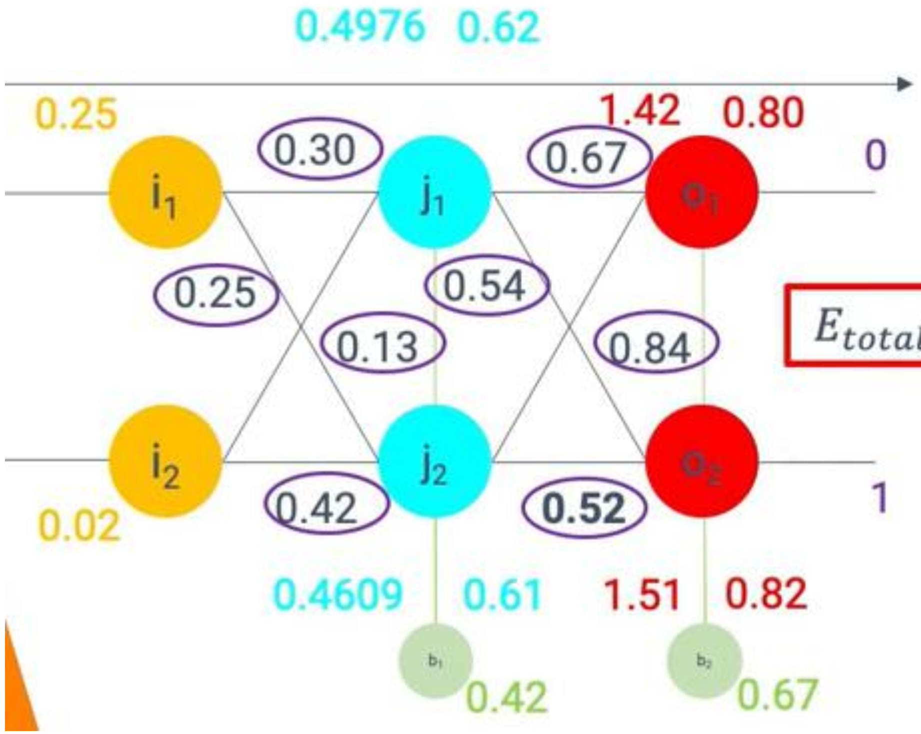


$$\frac{\partial E_{total}}{\partial w_8} = \frac{\partial E_{total}}{\partial out_{o2}} \cdot \boxed{\frac{\partial out_{o2}}{\partial in_{o2}}} \cdot \frac{\partial in_{o2}}{\partial w_8}$$

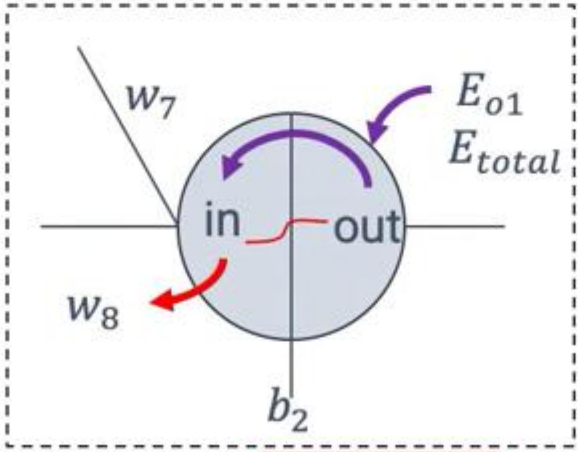
$$out_{o2} = \frac{1}{1 + e^{-in_{o2}}}$$

$$\frac{\partial out_{o2}}{\partial in_{o2}} = out_{o2}(1 - out_{o2}) = 0.82(1-0.82) = 0.1476$$

Backward propagation



$E_{total} = \mathbf{0.3362}$

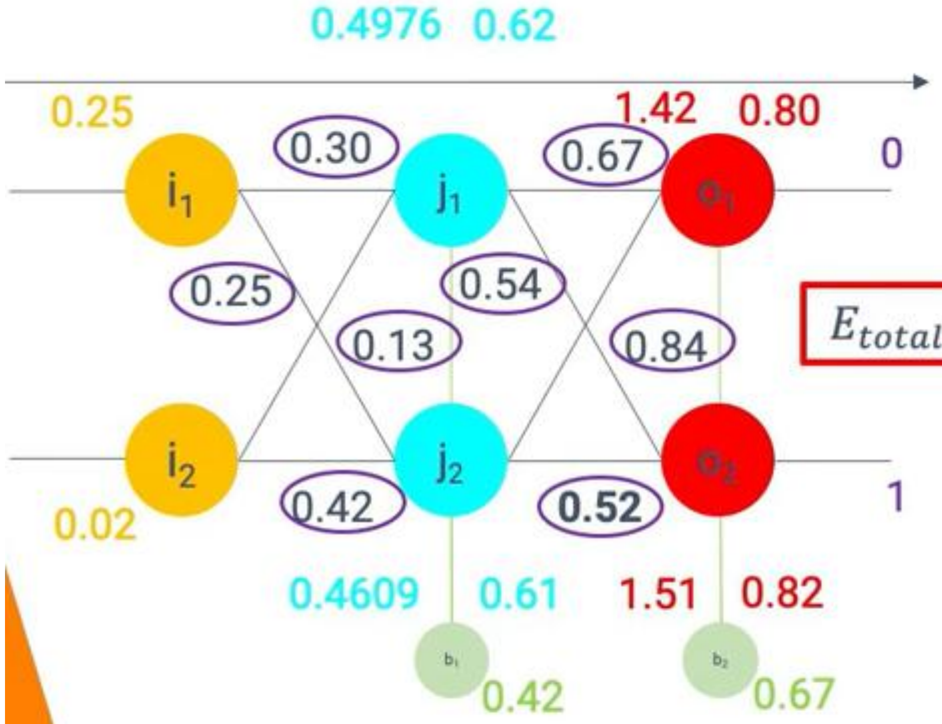


$$\frac{\partial E_{total}}{\partial w_8} = \frac{\partial E_{total}}{\partial out_{o2}} \cdot \frac{\partial out_{o2}}{\partial in_{o2}} \cdot \boxed{\frac{\partial in_{o2}}{\partial w_8}}$$

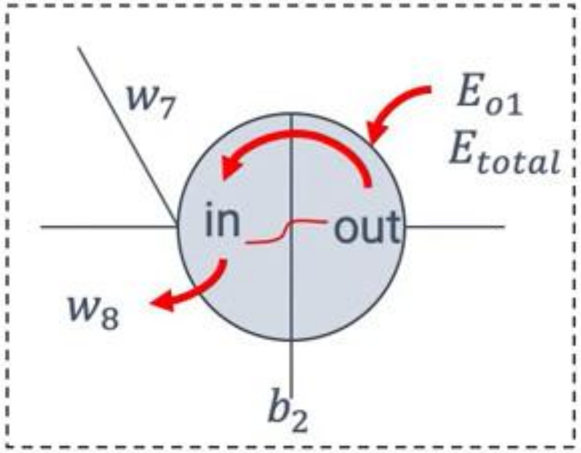
$$in_{o2} = w_8 \cdot out_{j2} + w_7 \cdot out_{j1} + b_2$$

$$\frac{\partial in_{o2}}{\partial w_8} = out_{j2} = 0.61$$

Backward propagation



$E_{total} = \mathbf{0.3362}$



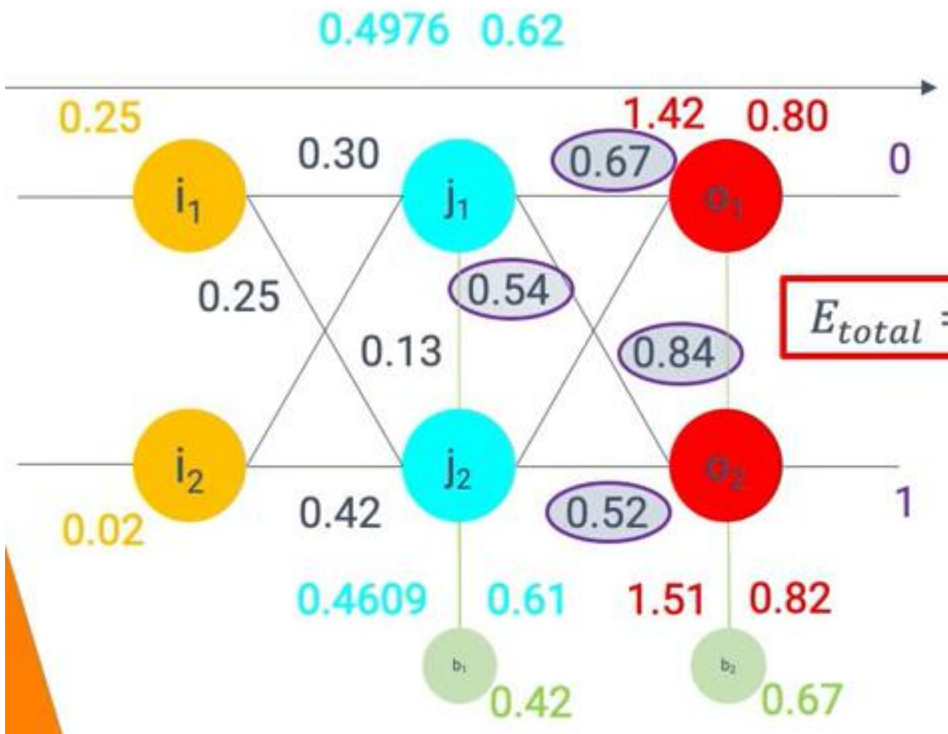
$$\frac{\partial E_{total}}{\partial w_8} = \frac{\partial E_{total}}{\partial out_{o2}} \cdot \frac{\partial out_{o2}}{\partial in_{o2}} \cdot \frac{\partial in_{o2}}{\partial w_8}$$

$$\frac{\partial E_{total}}{\partial w_8} = -0.18 \cdot 0.1476 \cdot 0.52 = -0.016$$

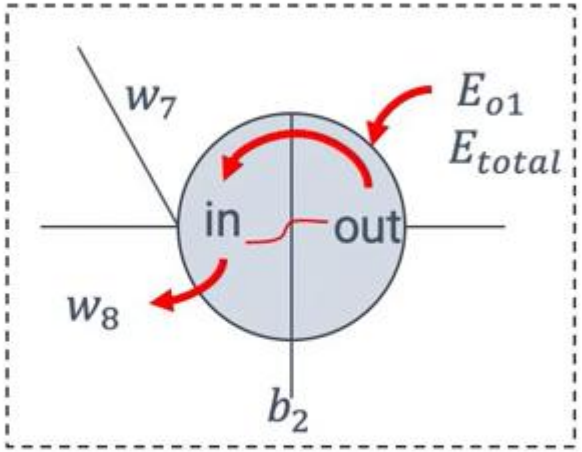
$$w_{8\_updated} = w_8 - \epsilon \frac{\partial E_{total}}{\partial w_8} = 0.52 - 0.8 \cdot -0.016 = \mathbf{0.5328}$$

Learning rate

Backward propagation



$E_{total} = \mathbf{0.3362}$

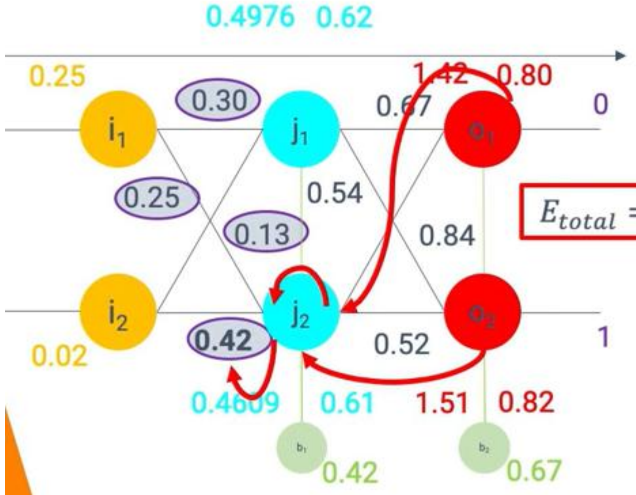


Adjust weights for output layer

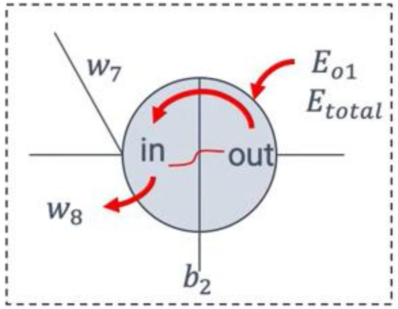
$$\frac{\partial E_{total}}{\partial w_8} = \frac{\partial E_{total}}{\partial out_{o2}} \cdot \frac{\partial out_{o2}}{\partial in_{o2}} \cdot \frac{\partial in_{o2}}{\partial w_8}$$

$$\frac{\partial E_{total}}{\partial w_8} = \underbrace{(out_{o2} - target_{o2}) \cdot out_{o2}(1 - out_{o2})}_{\frac{\partial E_{total}}{\partial out_{o2}} \cdot \frac{\partial out_{o2}}{\partial in_{o2}} = \frac{\partial E_{total}}{\partial in_{o2}} = \delta_{o2}} \cdot out_{j2}$$

Backward propagation



$\boxed{E_{total} = \mathbf{0.3362}}$



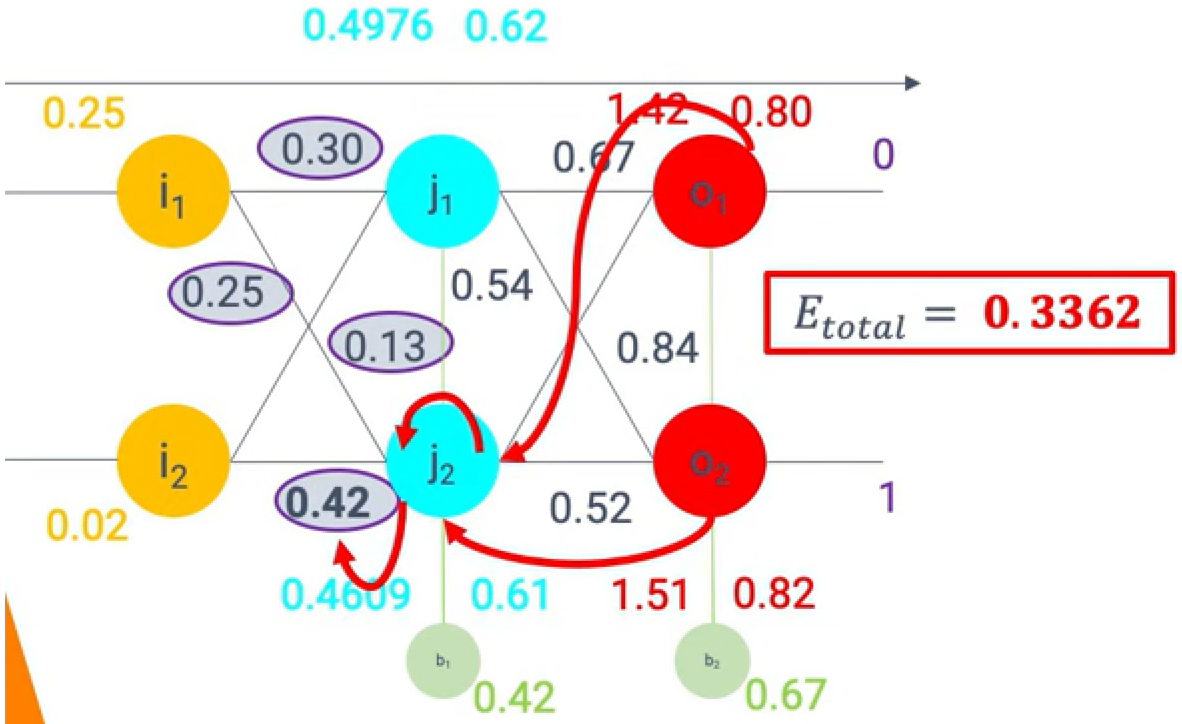
Adjust weights for hidden layer

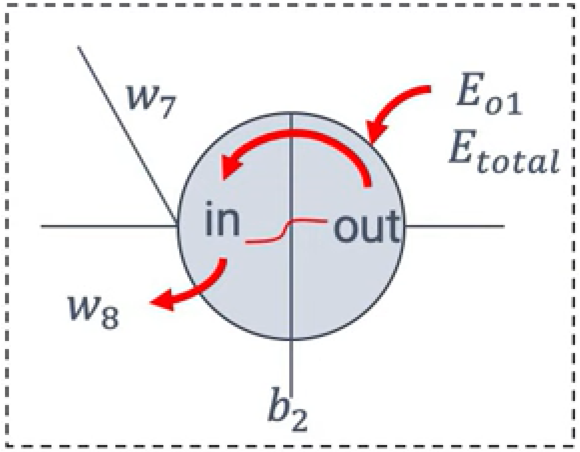
$$\frac{\partial E_{total}}{\partial w_4} = \boxed{\frac{\partial E_{total}}{\partial out_{j2}}} \cdot \frac{\partial out_{j2}}{\partial in_{j2}} \cdot \frac{\partial in_{j2}}{\partial w_4}$$

$$\frac{\partial E_{total}}{\partial out_{j2}} = \frac{\partial E_{o1}}{\partial out_{j2}} + \frac{\partial E_{o2}}{\partial out_{j2}}$$

$$\frac{\partial E_{o2}}{\partial out_{j2}} = \frac{\partial E_{o2}}{\partial in_{o2}} \cdot \frac{\partial in_{o2}}{\partial out_{j2}}$$

Backward propagation





$$\frac{\partial E_{total}}{\partial w_4} = \boxed{\frac{\partial E_{total}}{\partial out_{j2}}} \cdot \frac{\partial out_{j2}}{\partial in_{j2}} \cdot \frac{\partial in_{j2}}{\partial w_4}$$

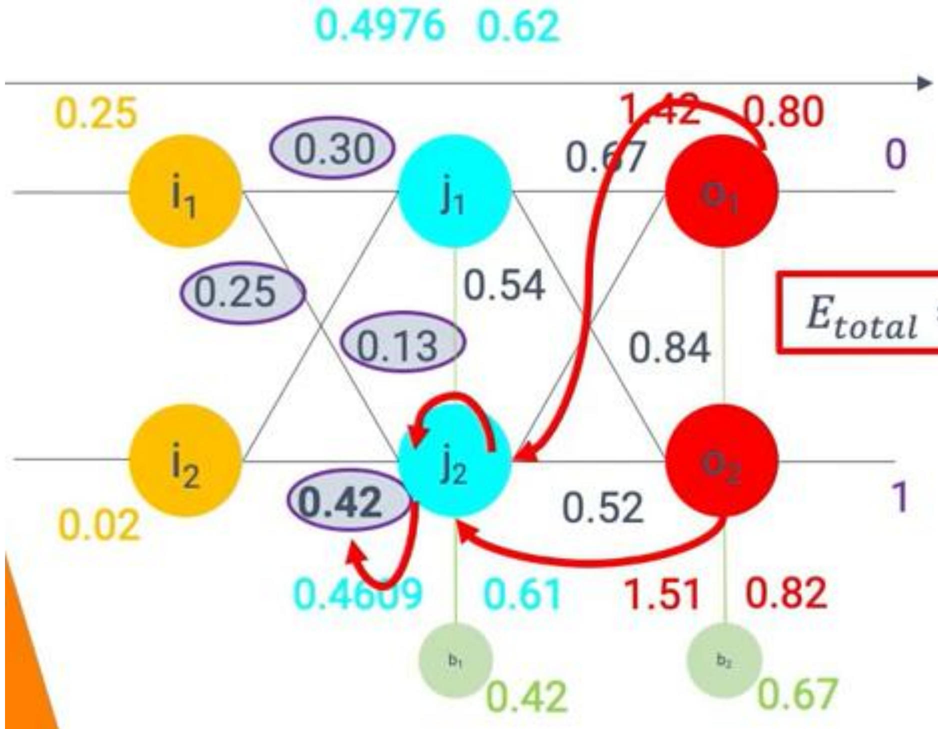
$$\frac{\partial E_{o2}}{\partial out_{j2}} = \frac{\partial E_{o2}}{\partial in_{o2}} \cdot \frac{\partial in_{o2}}{\partial out_{j2}}$$

$$\frac{\partial E_{o2}}{\partial in_{o2}} = \frac{\partial E_{o2}}{\partial out_{o2}} \cdot \frac{\partial out_{o2}}{\partial in_{o2}} = -0.18 \cdot 0.1476$$

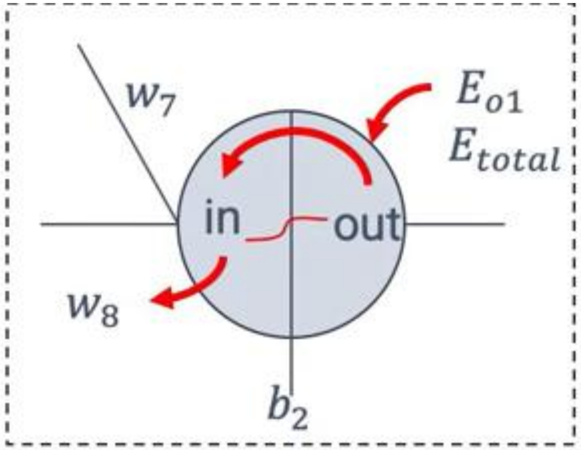
$$\frac{\partial E_{o2}}{\partial in_{o2}} = -0.0266$$

$$\frac{\partial E_{o2}}{\partial out_{j2}} = -0.0266 \cdot 0.52 = -0.0138$$

Backward propagation



$E_{total} = \mathbf{0.3362}$

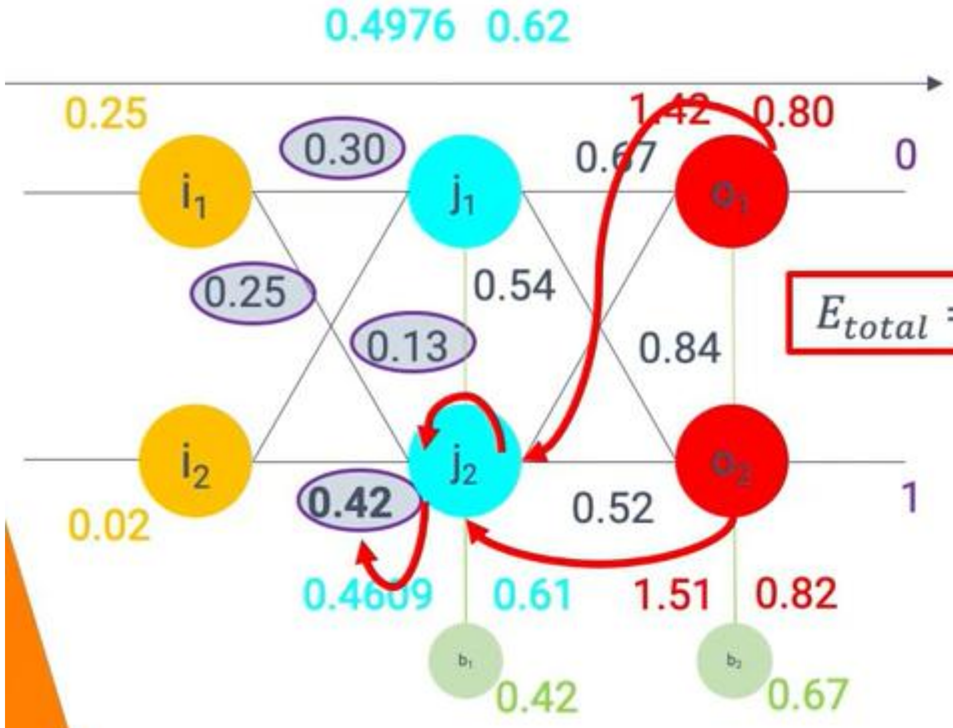


Adjust weights for the hidden layer

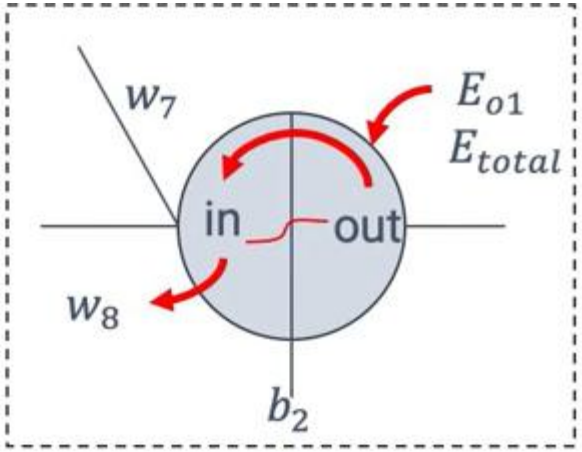
$$\frac{\partial E_{total}}{\partial w_4} = \frac{\partial E_{total}}{\partial out_{j2}} \cdot \boxed{\frac{\partial out_{j2}}{\partial in_{j2}}} \cdot \frac{\partial in_{j2}}{\partial w_4}$$

$$\frac{\partial out_{j2}}{\partial in_{j2}} = out_{j2}(1 - out_{j2}) = 0.16$$

Backward propagation



$E_{total} = \mathbf{0.3362}$



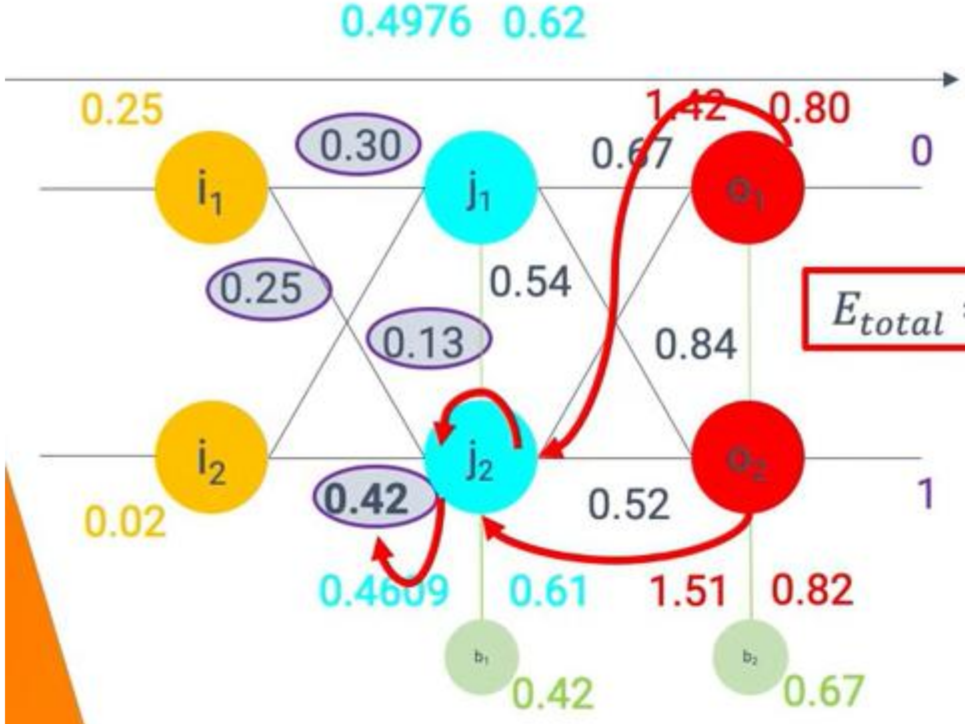
Adjust weights for the hidden layer

$$\frac{\partial E_{total}}{\partial w_4} = \frac{\partial E_{total}}{\partial out_{j2}} \cdot \frac{\partial out_{j2}}{\partial in_{j2}} \cdot \boxed{\frac{\partial in_{j2}}{\partial w_4}}$$

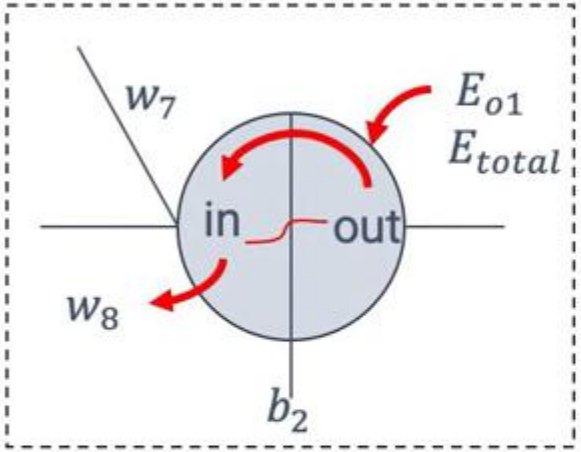
$$j_2 = i_1 \cdot w_3 + i_2 \cdot w_4 + b_1$$

$$\frac{\partial in_{j2}}{\partial w_4} = i_2 = 0.02$$

Backward propagation



$E_{total} = \mathbf{0.3362}$



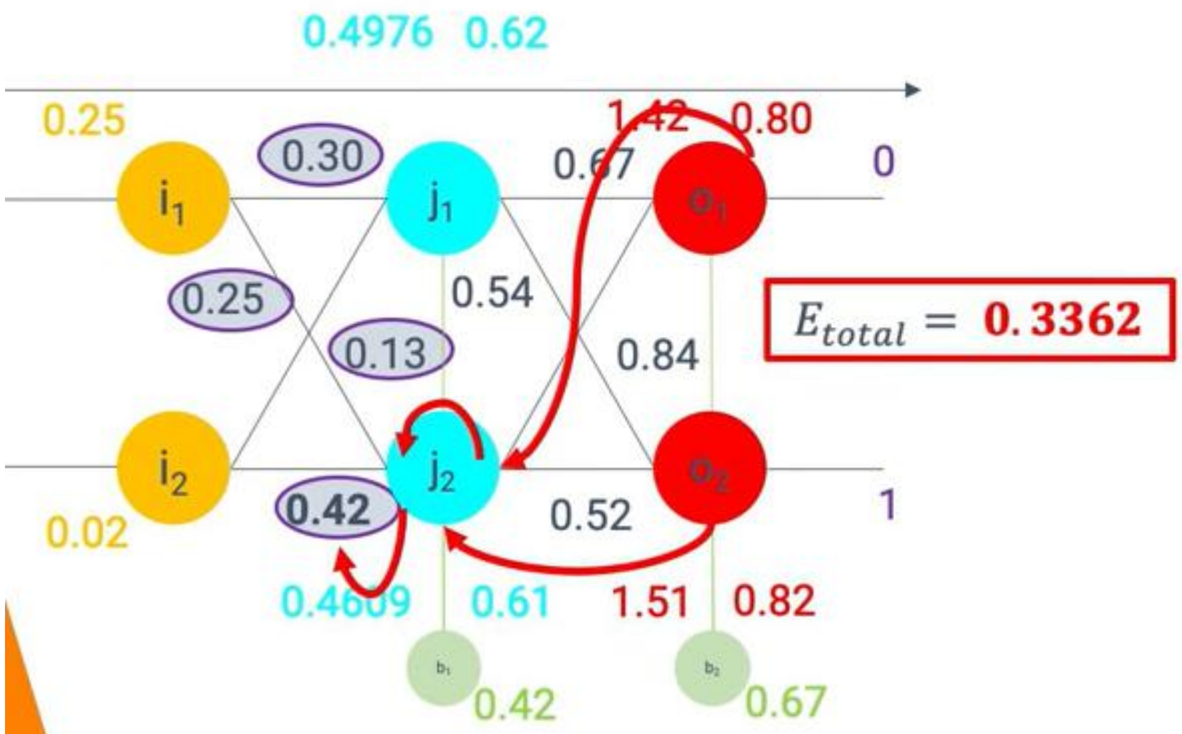
Adjust weights for the hidden layer

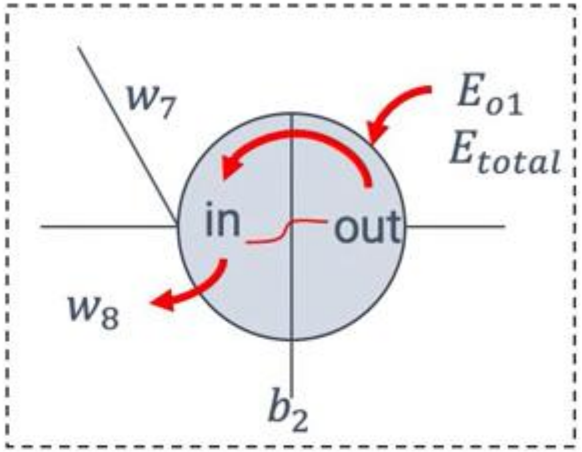
$$\frac{\partial E_{total}}{\partial w_4} = \frac{\partial E_{total}}{\partial out_{j2}} \cdot \frac{\partial out_{j2}}{\partial in_{j2}} \cdot \frac{\partial in_{j2}}{\partial w_4}$$

$$\frac{\partial E_{total}}{\partial w_4} = -0.0293 \cdot 0.16 \cdot 0.02$$

$$\frac{\partial E_{total}}{\partial w_4} = -0.00009376$$

Backward propagation





Adjust weights for the hidden layer

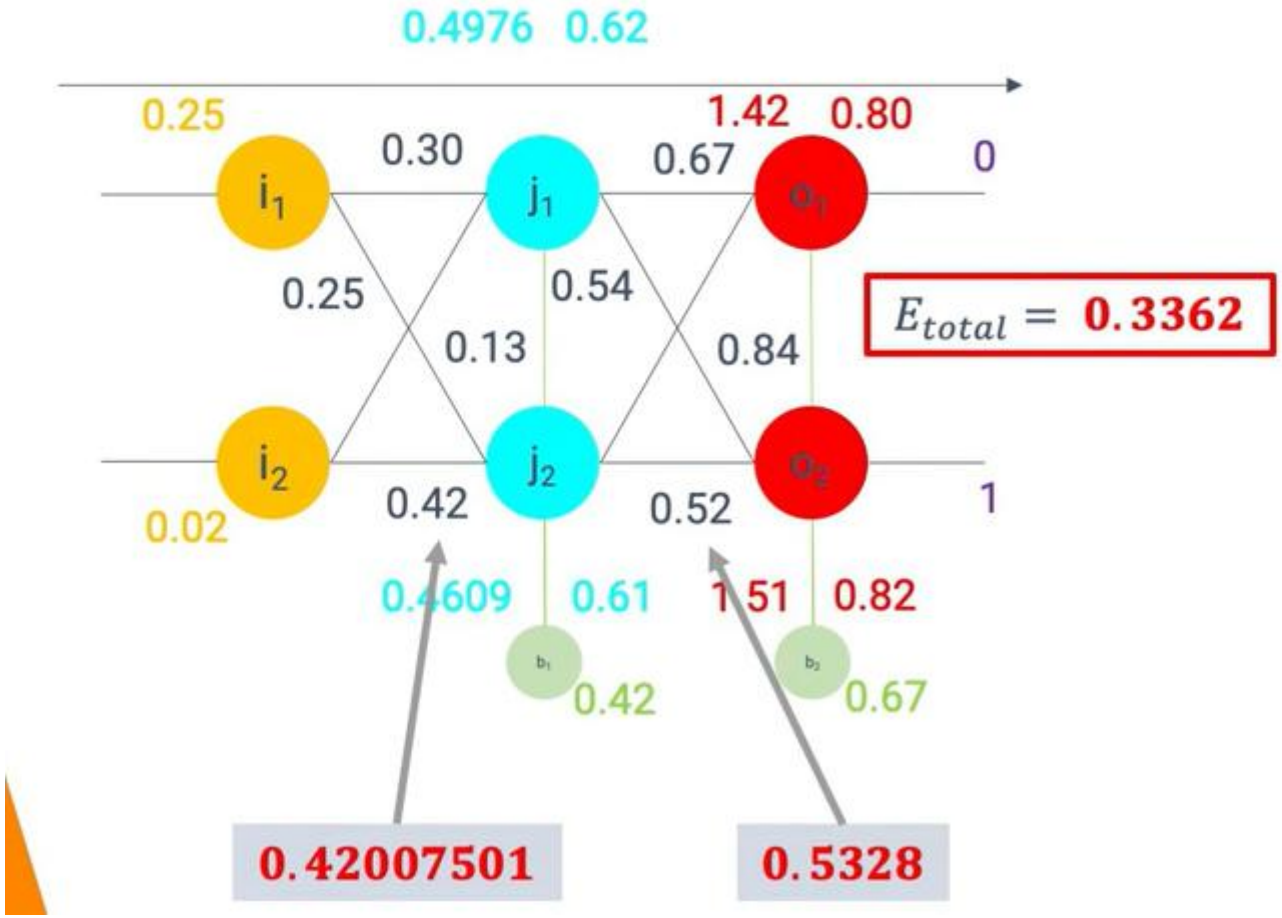
$$\frac{\partial E_{total}}{\partial w_4} = \frac{\partial E_{total}}{\partial out_{j2}} \cdot \frac{\partial out_{j2}}{\partial in_{j2}} \cdot \frac{\partial in_{j2}}{\partial w_4}$$

$$w_{4\_updated} = w_4 - \epsilon \frac{\partial E_{total}}{\partial w_4}$$

$$w_{4\_updated} = 0.42 - 0.8 \cdot -0.00009376$$

$$w_{4\_updated} = 0.42007501$$

Backward propagation



Summary

- When we fed forward the 0.05 and 0.1 inputs originally, the error on the network was 0.298371109.
- After this first round of backpropagation, the total error is now down to 0.291027924.
- It might not seem like much, but after repeating this process 10,000 times, for example, the error plummets to 0.0000351085.
- At this point, when we feed forward 0.05 and 0.1, the two outputs neurons generate 0.015912196 (vs 0.01 target) and 0.984065734 (vs 0.99 target).